

Track an Object in 3D Space

FP.1 Match 3D Objects

The first task was to match the bounding boxes between successive frames of the image sequence. This required iterating through the vector of bounding boxes in the previous frame in an outer loop, and then iterating through all of the bounding boxes in the current frame. The matched keypoints were then looped through and the corresponding keypoints were retrieved for the current matched keypoint pair. If the keypoint in the previous frame was enclosed by the current bounding box for the previous frame and the keypoint for the current frame was enclosed by the bounding box for the current frame, then a correspondence variable was increased by one. At the end of the loop through all matched keypoint pairs, the correspondence variable value is pushed onto a vector that stores the number of matched keypoints enclosed by this particular combination of bounding boxes from the previous and current frames. When all bounding boxes have been looped through for the current frame, the index of the max value of the correspondence integer vector is determined. That index is used to retrieve the bounding box and therefore its boxID from the vector of bounding boxes for the current frame. Finally the boxIDs for the bounding boxes of the previous frame and current frame are found and saved into an *std::map* structure for return.

FP.2 Compute LIDAR Based Time to Collision

The LIDAR points from the previous and current frames are used to calculate the time to collision. This is done by finding the minimum LIDAR point in the X direction relative to the sensor that also has a Y value within the ego lane width for each frame. The time between successive images in the image sequence is calculated by taking the inverse of the frame rate. These values are then used in the equation below to determine a time to collision in seconds.

$$TTC = \text{minXCurr} * dT / (\text{minXPrev} - \text{minXCurr})$$

FP.3 Associate Keypoint Correspondences with Bounding Boxes

For this task, all keypoints matches were iterated through and the corresponding keypoints for the previous and current frames were retrieved. If the bounding box enclosed both of these keypoints, then the keypoint match was pushed onto the back of the vector of keypoint matches for the *boundingBox* data structure.

FP.4 Compute Camera Based Time to Collision

By looping over all matched keypoints from beginning to end minus one in an outer loop and from beginning plus one to end in an inner loop the current and previous frame's keypoints for the inner and out loops can be retrieved and used to determine the distance by calculated the difference between these values. If the distance is greater than or equal to a minimum distance value set to 100.0 in this case, and the distance is greater than the *std::number_limits_epsilon* value, the distance ratio will be calculated and pushed onto the back of a vector of distance ratios. After all keypoint matched pairs have been iterated through both in the inner and outer loops the vector of *distRatios* is first sorted and then the size of this vector is then determined. If the vector has an odd number of elements, the element in the middle of the vector is set as the *medianDistRatio* and used in the formula below. If the vector has an even number of elements then the two middle elements are retrieved and used to calculate the average of these two elements and used as the *medianDistRatio*. The time between successive frames in the image sequence is calculated by taking the inverse of the *frameRate* and all of these values are then used to calculate the time to collision with the equation below.

$$TTC = -dT / (1.0 - medianDistRatio)$$

FP.5 Performance Evaluation 1

For the combination of using the FAST detector with the BRIEF descriptor, several frames stood out from the rest with the calculated time to collisions based off of LIDAR points. Frame number 9 jumps from about 9 seconds for the previous frame to approximately 18 seconds for the current. Also frame number 12 goes to a value of approximately -10 s. The first unexpected result could have been caused by an outlier LIDAR point. There is also the issue of assuming a constant velocity for this model where in reality not only the velocity but also the acceleration of both moving bodies change with respect to time. The solution to improve results could be to use a more accurate model of the physics of the moving bodies and/or to implement a better method of removing outlier LIDAR points from the preceding vehicle.

FP.6 Performance Evaluation 2

Every detector and descriptor combination was used to calculate the time to collision for each of the 18 frames and recorded in the table below. The values were then graphed where you can easily see the grouping of the effect of using various combinations on the outcome of the time to collision calculations based on the camera alone.

Camera Based Time to Collision (s)																			
Detector	Descriptor	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7	Frame 8	Frame 9	Frame 10	Frame 11	Frame 12	Frame 13	Frame 14	Frame 15	Frame 16	Frame 17	Frame 18
Sh-Tomas	BRISK	13.90	12.99	13.49	12.39	12.66	13.25	12.72	13.71	11.49	13.46	11.30	11.65	11.72	11.51	9.34	10.46	11.40	9.12
	BRIEF	13.89	13.15	11.40	13.47	12.18	13.09	16.34	11.86	11.60	13.94	11.78	11.62	11.69	11.03	11.30	11.10	11.19	8.15
	ORB	13.50	12.67	11.40	12.06	12.30	13.42	11.97	12.18	11.27	13.13	11.30	11.48	12.36	11.34	10.26	10.87	10.73	8.56
Sh-Tomas	FREAK	13.65	13.09	11.04	12.77	12.48	13.91	12.87	12.85	11.60	13.20	11.33	11.79	12.25	11.51	10.09	11.43	10.87	8.23
	BRISK	10.91	10.20	-80.85	11.40	89.83	12.99	11.66	13.57	nan	-0.20	11.67	11.39	204.16	5.66	-29.56	6.65	12.50	-inf
	BRIEF	10.91	11.01	-11.47	11.40	34.75	14.70	14.27	13.57	2.86	-inf	11.67	11.39	13.36	5.66	-29.56	6.76	11.60	-0.20
Harris	ORB	10.91	11.01	-11.47	11.40	39.50	13.62	13.50	13.57	nan	10.29	11.67	11.39	13.36	5.66	-29.56	6.72	12.50	-inf
	FREAK	8.75	10.20	-10.30	11.58	35.38	13.66	12.28	12.64	nan	10.29	11.67	11.39	13.36	5.66	-29.56	6.69	12.50	-inf
	BRISK	12.42	12.32	12.21	12.55	13.63	13.99	13.84	12.30	12.37	13.89	12.00	12.11	11.91	11.40	11.07	11.13	10.27	11.65
FAST	BRIEF	11.57	12.31	13.64	12.63	13.15	13.46	13.16	12.51	12.37	13.89	11.68	11.76	11.71	10.45	11.03	10.32	9.68	11.55
	ORB	11.80	12.26	13.06	12.63	13.15	12.46	13.40	12.69	12.37	13.87	11.68	11.76	11.72	10.69	10.36	10.63	9.45	11.53
	FREAK	11.88	12.93	13.06	13.42	12.29	11.66	12.65	12.62	12.30	12.70	12.12	12.20	11.62	11.12	10.36	10.63	10.10	11.51
BRISK	BRISK	13.03	12.63	11.63	15.20	26.43	16.62	15.60	16.31	13.90	13.09	12.27	11.61	11.85	12.04	12.47	11.30	9.29	10.54
	BRIEF	12.82	16.86	11.66	20.62	19.71	16.16	15.30	18.42	15.46	11.16	13.36	13.69	11.87	10.91	11.06	11.10	10.92	10.47
	ORB	14.68	19.62	13.27	16.37	20.30	18.80	15.52	16.07	15.08	11.26	13.66	11.02	11.52	11.98	11.47	11.29	9.98	11.45
BRISK	FREAK	12.36	23.53	13.78	13.64	23.07	14.82	15.42	18.77	15.39	13.88	12.37	11.06	11.49	11.40	13.05	13.28	10.29	10.38
	BRISK	16.51	13.10	12.42	29.39	156.22	10.84	13.69	10.70	13.09	17.12	7.66	-inf	7.71	9.47	8.73	13.28	12.37	27.51
	BRIEF	20.66	47.69	102.10	13.90	22.85	15.13	55.20	22.78	70.71	16.26	17.26	13.79	13.30	9.71	8.82	10.98	14.15	18.90
ORB	ORB	87.33	10.12	16.64	26.93	32.31	18.45	215.41	9.39	21.49	16.86	8.29	35.64	9.69	5.66	12.85	11.35	11.38	29.89
	FREAK	12.18	20.12	12.23	14.42	743.24	11.54	-inf	8.99	13.31	-inf	7.36	35.65	6.53	57.39	8.59	7.77	9.84	16.87
	BRISK	11.90	15.66	13.37	14.85	14.75	14.71	15.88	13.92	13.82	11.87	12.36	10.57	10.09	10.04	10.00	10.38	9.64	8.83
AKAZE	BRIEF	13.28	14.91	13.67	13.85	14.55	13.91	15.93	14.11	13.51	11.59	12.36	11.57	10.10	9.86	9.27	9.70	9.26	9.15
	ORB	12.40	14.85	13.28	14.36	15.81	13.83	15.41	13.97	13.45	11.56	12.14	11.22	10.08	10.35	9.89	9.65	9.07	9.08
	FREAK	12.12	13.88	13.94	14.18	15.84	14.44	15.37	14.07	13.67	11.78	12.16	10.77	10.78	9.88	9.63	10.35	9.15	8.79
AKAZE	AKAZE	12.34	13.94	12.88	14.25	16.06	13.87	15.34	14.14	13.84	11.55	12.13	10.92	11.08	10.58	9.55	9.75	9.06	8.97
	SIFT	11.59	13.03	13.54	18.94	14.99	11.57	14.05	15.00	12.94	10.81	12.76	9.81	9.60	9.27	9.90	9.07	8.95	8.83
	BRIEF	12.25	12.73	15.09	21.31	13.16	11.68	14.18	15.26	12.54	10.31	12.61	10.14	9.22	9.27	9.90	9.07	8.95	8.83
SIFT	FREAK	11.24	13.62	13.54	20.49	14.96	11.97	14.41	13.17	13.51	11.36	12.06	10.09	10.06	9.76	8.99	8.72	9.20	9.39

